

Community weighted traits

Prabu Ravindran

Plots, coverages, and data frames

We will demonstrate one-level and two-level community weighting with a single plot. Table 1a shows, a plot for the one-level example - there is a single subplot for which plot composition is reported. For the two-level example, Table 1b shows a plot with *two* subplots - in general there can be *two or more* subplots per plot. The two-level refers to the fact that weighted traits are first computed at the subplot level and these subplot traits are weighted using subplot weights to produce a plot level trait value. One-level weighting is a special case of two-level weighting, where there is one subplot. Table 1, provides the employed notation. A, B, C, D represent the cover classes (e.g. species, plant functional types). In Tables 1a, 1b, A_r^s represents the r -th replicate for cover class A in the s -th subplot. In Table 1b, there are 2 replicates for cover class A in subplot 1 (A_1^1, A_2^1) and 2 replicates for D in subplot 2 (D_1^1, D_2^1). The cover weight (**CoverWeight** column) for the various cover classes for the subplots are reported in Tables 1c, 1d. c_k^s , represents the coverage weight for cover class k in subplot s . Note that we include subplot identifier in the superscript!

Table 1: Plots and coverages.

(a) Plot for **one-level** weighting.

A_1^1	A_2^1
B_1^1	E_1^1

(b) Plot for **two-level** weighting.

A_1^1	A_2^1	A_1^2	E_1^2
B_1^1	E_1^1	D_1^2	D_2^2

(c) Coverage for **one-level** weighting

SubplotID	CoverID	CoverWeight
1	A	c_A^1
1	B	c_B^1
1	E	c_E^1

(d) Coverage for **two-level** weighting

SubplotID	CoverID	CoverWeight
1	A	c_A^1
1	B	c_B^1
1	E	c_E^1
2	A	c_A^2
2	E	c_E^2
2	D	c_D^2

The data frames for the one-level and two level cases are shown in Table 2. The trait information for **CoverID**= k , replicate r for **SubplotID**= s is provided as: (**TraitMean**, **TraitSdev**)= $(\mu_{k_r}^s, \sigma_{k_r}^s)$. If the replicates for **CoverID** are reduced we get Table 2b, where the reduced value in **TraitSdev** is the root-mean-square of the standard deviations for the **CoverID**. After reduction, the situation becomes equivalent to the case of having one replicate per **CoverID**!

Table 2: Data frames.

(a) Data frame for **one-level** weighting.

SubplotID	SubplotWeight	CoverID	CoverWeight	TraitMean	TraitSdev
1	w^1	A_1	c_A^1	$\mu_{A_1}^1$	$\sigma_{A_1}^1$
1	w^1	A_2	c_A^1	$\mu_{A_2}^1$	$\sigma_{A_2}^1$
1	w^1	B_1	c_B^1	$\mu_{B_1}^1$	$\sigma_{B_1}^1$
1	w^1	E_1	c_E^1	$\mu_{E_1}^1$	$\sigma_{E_1}^1$

(b) Reduced data frame for **one-level** weighting.

SubplotID	SubplotWeight	CoverID	CoverWeight	TraitMean	TraitSdev
1	w^1	A_1	c_A^1	$\frac{\mu_{A_1}^1 + \mu_{A_1}^2}{2}$	$\sqrt{\frac{(\sigma_{A_1}^1)^2 + (\sigma_{A_2}^1)^2}{2}}$
1	w^1	B_1	c_B^1	$\mu_{B_1}^1$	$\sigma_{B_1}^1$
1	w^1	E_1	c_E^1	$\mu_{E_1}^1$	$\sigma_{E_1}^1$

(c) Data frame for **two-level** weighting

SubplotID	SubplotWeight	CoverID	CoverWeight	TraitMean	TraitSdev
1	w^1	A_1	c_A^1	$\mu_{A_1}^1$	$\sigma_{A_1}^1$
1	w^1	A_2	c_A^1	$\mu_{A_2}^1$	$\sigma_{A_2}^1$
1	w^1	B_1	c_B^1	$\mu_{B_1}^1$	$\sigma_{B_1}^1$
1	w^1	E_1	c_E^1	$\mu_{E_1}^1$	$\sigma_{E_1}^1$
2	w^2	A_1	c_A^2	$\mu_{A_1}^2$	$\sigma_{A_1}^2$
2	w^2	E_1	c_E^2	$\mu_{E_1}^2$	$\sigma_{E_1}^2$
2	w^2	D_1	c_D^2	$\mu_{D_1}^2$	$\sigma_{D_1}^2$
2	w^2	D_2	c_D^2	$\mu_{D_2}^2$	$\sigma_{D_2}^2$

Community weighting and sampling

The following notation will be used:

- $[\widetilde{c}_k^s]_i$: i -th sample from a specified distribution $D_c(c_k^s)$.
- $[\widetilde{t}_{k_r}^s]_i$: i -th sample from a specified distribution $D_t(\mu_{k_r}^s, \sigma_{k_r}^s)$.

The community weighted trait for a plot can be obtained as follows:

- For $i \in 1, \dots, M$:
 - For each subplot s :
 - * For every **CoverID** = k :
 - Choose: r (if replicates exist for **CoverID**).
 - Sample: $([\widetilde{c}_k^s]_i, [\widetilde{t}_{k_r}^s]_i)$.
 - * Compute: $[t^s]_i = \frac{\sum_k [\widetilde{c}_k^s]_i [\widetilde{t}_{k_r}^s]_i}{\sum_k [\widetilde{c}_k^s]_i}$ (subplot level weighting).
 - Compute $\widetilde{y}_i = \frac{\sum_s w^s [t^s]_i}{\sum_s w^s}$ (plot level weighting).
- Compute:
 - $\widehat{\mu} = \text{mean}(\widetilde{y}_1, \dots, \widetilde{y}_M)$ (community weighted mean trait for plot),
 - $\widehat{\sigma} = \text{sdev}(\widetilde{y}_1, \dots, \widetilde{y}_M)$ (community weighted standard deviation of trait for plot).

Example (one-level case): Let $A_{r=2}$ was chosen. Then,

$$[t^1]_i = \frac{[\widetilde{c}_A^1]_i [\widetilde{t}_{A_2}^1]_i + [\widetilde{c}_B^1]_i [\widetilde{t}_{B_1}^1]_i + [\widetilde{c}_E^1]_i [\widetilde{t}_{E_1}^1]_i}{[\widetilde{c}_A^1]_i + [\widetilde{c}_B^1]_i + [\widetilde{c}_E^1]_i}$$

,

$$\widetilde{y}_i = \frac{w^1 [t^1]_i}{w^1}$$

Example (two-level case): Let $A_{r=1}$ and $D_{r=2}$ was chosen for subplots $s = 1$ and $s = 2$ respectively. Then,

$$[t^1]_i = \frac{[\widetilde{c}_A^1]_i [\widetilde{t}_{A_1}^1]_i + [\widetilde{c}_B^1]_i [\widetilde{t}_{B_1}^1]_i + [\widetilde{c}_E^1]_i [\widetilde{t}_{E_1}^1]_i}{[\widetilde{c}_A^1]_i + [\widetilde{c}_B^1]_i + [\widetilde{c}_E^1]_i}$$

,

$$[t^2]_i = \frac{[\widetilde{c}_A^2]_i [\widetilde{t}_{A_1}^2]_i + [\widetilde{c}_E^2]_i [\widetilde{t}_{E_1}^2]_i + [\widetilde{c}_D^2]_i [\widetilde{t}_{D_2}^2]_i}{[\widetilde{c}_A^2]_i + [\widetilde{c}_E^2]_i + [\widetilde{c}_D^2]_i}$$

,

$$\widetilde{y}_i = \frac{w^1 [t^1]_i + w^2 [t^2]_i}{w^1 + w^2}$$

Distributions for sampling

Trait sampling

For generating the samples $[\widetilde{t_{k_r}^s}]_i$, the distribution D_t is a truncated normal with mean and standard deviation: $(\mu_{k_r}^s, \sigma_{k_r}^s)$. Along with the two moments, the truncation support needs specification. Additionally, value clipping of the generated samples may be required (e.g. disallowing non-negative values).

CoverWeight sampling

- **Percent cover:** If **CoverWeight** = p , $p \in [l_j, r_j]$, a **TrunNormal** distribution with mean p , standard deviation d_j and truncation to one d_j is used for sampling $[\widetilde{c_k^s}]_i$. Additionally, the samples are clipped to the range $[0, 100]$

Table 3: Specification for percent cover

Index	Left	Right	Sdev
1	l_1	r_1	d_1
...	...	...	...
B	l_B	r_B	d_B

- **Ordinal class:** If **CoverWeight** = j , a **Beta** distribution with mode p_j , concentration q_j , and support $[l_j, r_j]$ is used for sampling $[\widetilde{c_k^s}]_i$. The bin details are provided as in Table 4.

Table 4: Specification for ordinal cover

Index	Peak	Concentration	Left	Right
1	p_1	q_1	l_1	r_1
...	...	...	...	...
B	p_B	q_B	l_B	r_B

- **Role of Left, Right:**
 - For percent cover, **Left**, **Right** are used for specifying an uncertainty that depends on the range in which the percent cover lies.
 - For ordinal class specification, **Left**, **Right** specify the support of **Beta** distribution for that bin.